

RESEARCH ARTICLE

Annual extent of prescribed burning on moorland in Great Britain and overlap with ecosystem services

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Abstract

In the UK uplands, prescribed burning of unenclosed heath, grass and blanket bog ('moorland') is used to support game shooting and grazing. Burning on moorland is contentious due to its impact on peat soils, hydrology and habitat condition. There is little information on spatial and temporal patterns of burning, the overlap with soil carbon and sensitive habitats and, importantly, whether these patterns are changing. This information is required to assess the sustainability of burning and the effectiveness of new legislation. We developed a method for semi-automated detection of burning using satellite imagery – our best performing model has a balanced accuracy of 84.9%. We identified annual burn areas in Great Britain in five burning seasons from 2017/18 to 2021/22 of 8333 to 20 974 ha (average 15 250 ha year⁻¹). Annual extent in England in 2021/22 was 73% lower than the average of the four previous seasons. Burning was identified over carbon-rich soils (mean 5150 ha or 34% by area of all burning annually) and on steep slopes – 915 ha across the five seasons (1.3%), contravening guidance. Burning (>1 ha) was recorded in 14% of UK protected areas (PAs) and, within these, the percentage area of moorland burned varied from 2 to 31%. In England in some years, the percentage area of moorland burned inside PAs was higher than outside, while this was not the case in Scotland. Burning in sensitive alpine habitats totalled 158 ha across the five seasons. The reduction in burned area in England in 2021/22 could relate to England-specific legislation, introduced in May 2021, to prohibit burning on deep peat in PAs. This suggests that regulation can be effective. However, the continued overlap with sensitive features suggests that burning falls short of sustainable practices. Our method will enable repeatable re-assessment of burning extents and overlap with ecosystem services.

Introduction

Large areas of the UK uplands comprise unenclosed heath, grass and blanket bog vegetation ('moorland'). This semi-natural habitat is largely of anthropogenic origin, formed by centuries of deforestation, grazing and burning (Thompson et al., 1995). Moorlands nonetheless provide various ecosystem services: (i) they often overlie carbon-rich (peat) soils, a significant terrestrial carbon store with over 3.2 billion tonnes found in blanket bog alone (Bain et al., 2011); (ii) the uplands supply >70% of the UK's drinking water (Watts et al., 2001) and regulate downstream flow regimes (Gao et al., 2017; Robinson

et al., 2013); and (iii) they are important for biodiversity, supporting habitats and species of conservation importance (Pearce-Higgins et al., 2009; Thompson et al., 1995) with large areas afforded protected area (PA) status (JNCC, 2022a, 2022b). UK moorland should therefore make a crucial contribution to mitigating the climate and nature emergencies. Many moorlands are, however, degraded by factors including atmospheric pollution, inappropriate grazing by domestic livestock and/or native herbivores, drainage and fire (Glaves et al., 2013; Holden et al., 2007; Sansom, 1999).

The use of fire is widespread in the UK uplands. This prescribed burning mainly takes place to stimulate new

moorland vegetation growth, to support recreational shooting of Red Grouse *Lagopus lagopus scotica*, with other uses including grazing management for livestock and deer (GWCT, 2023a; Scottish Government, 2021a). Burning for Red Grouse involves strip burning moorland vegetation on ~8–25 year rotations to create a mosaic of ages and heights (Thacker et al., 2015), and extends across ~8550 km² (Douglas et al., 2015), ~9% of the UK uplands [assuming that ~40% of the UK is upland (RSPB, 2007)].

Evidence reviews identify positive and negative impacts of moorland vegetation burning on ecosystem services including carbon sequestration, water quality, peatland hydrology, habitat quality and biodiversity (e.g. Brown et al., 2015; Glaves et al., 2013; Harper et al., 2018; Holland et al., 2022; Scottish Government, 2019; Thompson et al., 2016; Worrall et al., 2010). Overall, the weight of evidence is considered to show burning as damaging, especially over peat soils, for metrics including key peatland plants, peatland ecosystem health and the sustainability of peatland soils (DEFRA, 2021; IUCN UK Peatland Programme, 2023), and a government advisory body has called for an immediate ban on rotational burning on peatland (Climate Change Committee, 2020). Legislative change to ban burning on peat >40 cm deep within PAs (except under licence) in England was introduced on 1 May 2021 (UK Government, 2021), and similar regulation in Scotland is proposed (Scottish Parliament, 2023).

To understand the sustainability of burning, and the effect of regulatory changes on burning practices, it is

essential to map where and when burning occurs, and its overlap with ecosystem services impacted by burning. Previous moorland burn maps (e.g. Allen et al., 2016; Anderson et al., 2009; Douglas et al., 2015; Matthews et al., 2020; Yallop et al., 2006) have had limitations, because manual, visual image interpretation is time consuming, prohibiting repeat mapping and assessment of temporal change. Recent advances in Earth Observation data availability and tools provide the potential to automate this (e.g. Spracklen & Spracklen, 2023, although this was limited to Scotland).

We developed a method for semi-automated detection of moorland burning across Great Britain, to address evidence gaps around spatial and temporal patterns. We quantify annual burn extents and their overlap with the UK PA network, other sensitive habitats and peat soils.

Materials and Methods

Our workflow included data annotation, satellite imagery analysis/manipulation, model development followed by classification and map accuracy evaluation (Fig. 1). Prescribed burning takes place within defined autumn to spring 'burn seasons' (SM 1.1) allowing mapping of the appearance of characteristic burn scars in vegetation annually. We focussed on five burn seasons with available high-resolution (10 m) Sentinel-2 imagery, 2017/18 to 2021/22. Spatial data manipulation was undertaken using Google Earth Engine (GEE; Gorelick et al., 2017) and R 4.1.2 (R Core Team, 2022) using the packages sf

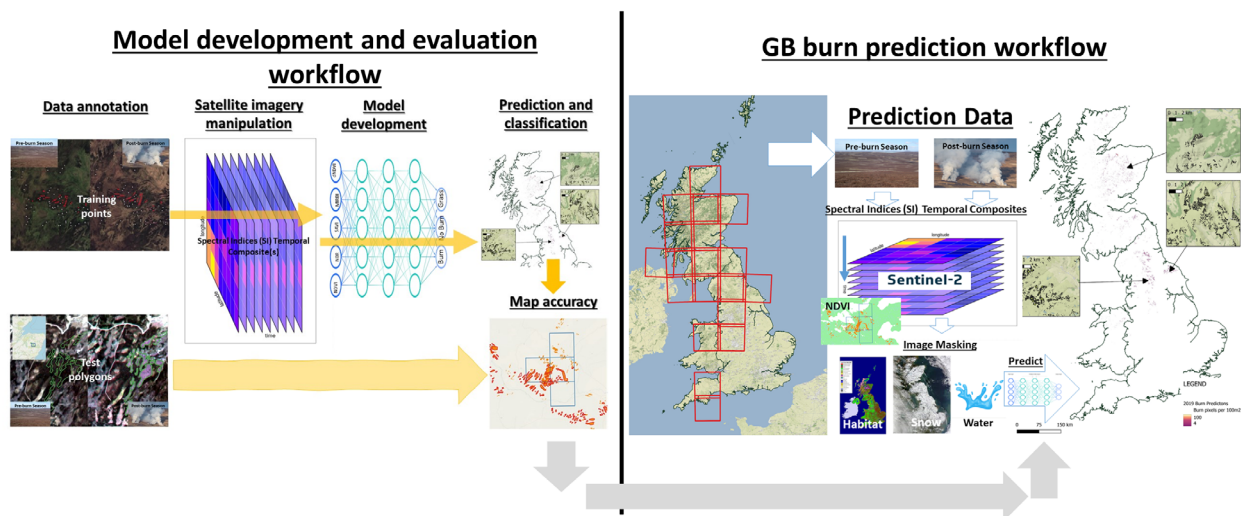


Figure 1. Method development stages for detection and predictive mapping of moorland burning. Broad prediction area of interest within the UK shown by red squares – see Figure S4. Sentinel-2 100 km² tiles (red squares) were used to filter imagery in Google Earth Engine to tile areas of interest and generate spectral indices temporal composite imagery prior to masking with habitat data, a snow mask and water mask and passed to trained deep neural network models for prediction.

(Pebesma, 2018) and terra (Hijmans et al., 2022) in RStudio (RStudio Team, 2022).

Data annotation

Training data

We created annotated burn and no-burn point datasets by selecting 10-m pixels located within burn scars and outside of burn scars (but within moorland habitats) in known burning areas (Douglas et al., 2015) from the 2018/2019 and 2019/2020 burn seasons. Using Sentinel-2 (Red, Green and Blue [RGB] bands only) imagery in GEE, burn scars were visually identified (e.g. Allen et al., 2016; Douglas et al., 2015; Matthews et al., 2020; Yallop et al., 2006). Spatial points generated ($n = 38\,392$) were used to extract information on spectral indices (SI) from training imagery for model development. See SM 1.1 for details.

Test data

We used a test dataset of 13, 1-km² for each prediction season to assess prediction accuracy (SM 1.2). We manually drew polygons around new burn scars and generated no-burn patches (SM 1.2), using Sentinel-2 (RGB) imagery in GEE. Polygon test datasets were selected to maximise the number of test pixels for evaluation.

Satellite image analysis and manipulation

Training imagery

Our analysis was based on Sentinel-2 satellite optical remote sensing data (https://sentinels.copernicus.eu/documents/247904/1848117/Sentinel-2_Data_Products_and_Access), with a spatial resolution of 10 or 20 m, but reprojected to a spatial resolution of 10 m using nearest neighbour resampling by GEE. Using the Harmonized Sentinel-2 MSI: MultiSpectral Instrument, Level-2A dataset in GEE, median temporal composite SI images (see SM 1.3; Table S2) were generated for spring (May–June) and summer (July–September), plus whole-year Soil Adjusted Vegetation Index variation (SAVIvar), an SI which helps distinguish deciduous grassland habitats (Lucas et al., 2011) – a potential source of false positives. SI's were generated for the area of interest (AOI; SM 1.1) in 2018, 2019 and 2020. Between-year median SI difference (dSI) was calculated from images for spring and summer of each year (SI's useful for detecting burn severity; Murphy et al., 2008), followed by cloud and shadow masking using the S2cloudless approach (see SM 1.3). This provided pre-fire and post-fire SI imagery for burn seasons 2018/19 and 2019/20 for training data extraction using our annotated point dataset.

Prediction imagery

Prediction images were prepared as above, for each burn season between 2017/18 and 2021/22, within defined upland areas of the UK (SM 1.4). Non-moorland habitats and areas of water and snow were masked out using UKCEH 2020 Land Cover Map (LCM) (Morton et al., 2021) and established water and snow masking approaches (SM 1.4; Fig. 1). The proportion of moorland pixels available for classification in the prediction area (Fig. 1) varied annually due, for example, to snow extent, but on average 75.8% of pixels contained valid data (SM 1.4; Table S4).

Classification model development

The annotated point SI and dSI data from the 2018/19 burn season were used to train deep neural network (DNN) models using TensorFlow (Abadi et al., 2016) via Python. Models were fitted using spring-only SI/dSI, summer-only SI/dSI and both spring and summer SI/dSI, and with/without SAVIvar, yielding six models (SM 2.1).

Classification and map accuracy evaluation

Model outputs were evaluated using polygon test datasets and binary transformed burn/no-burn probability predictions. Probabilities were transformed using a threshold that maximised the F_β -score ($\beta = 0.5$) when tested on the 2019/20 SI/dSI data. The F_β score ($\beta = 0.5$) minimises false positives in imbalanced classification (Brownlee, 2020).

Binary classifications were used to generate confusion matrices for each model and overall accuracy, balanced accuracy, false positive rate (FPR) and false negative rate (FNR) (Raschka, 2014; Story & Congalton, 1986). FPR is the proportion of test pixels falsely identified as a burn; FNR is the proportion of burn test pixels identified as no-burn.

Matthew's correlation coefficient (MCC) was calculated for the two top performing models based on accuracy, FPR and FNR. MCC is a correlation coefficient between observed and predicted classification, where higher values (closer to 1) are better, and is considered a balanced measure where classes are imbalanced (Boughorbel et al., 2017). Models were compared across all metrics and the top model (best performing on accuracy, MCC and FPR) selected for predicted burn area generation.

Moorland cutting

Mechanical cutting of moorland vegetation sometimes replaces burning (Douglas et al., 2017; Heinemeyer et al., 2019) and could also produce scars with differing

SI between years. We obtained polygons of 317 ha of cuts from northern England (SM 1.6) and assessed our models' ability to distinguish burns and cuts (SM 1.6).

Burning extent

Burn area was calculated from top model probability predictions converted to classified binary burn/no-burn pixels (F_{β} -Score threshold), summed and converted to area based on pixel size. Country-specific (Scotland, England and Wales) extents were calculated using national boundaries (ONS, 2022). 95% confidence intervals were calculated using stratified estimation (Olofsson et al., 2013). Burn areas were also calculated for only those pixels useable in all years, that is, not obscured by cloud or snow, but because extents were similar (SM 2.2), we report all data extents below.

Overlap of burning with ecosystem services

Burned area overlap with ecosystem services was assessed using spatial layers as described below.

Soil carbon

Intersection of burning with peat soils was undertaken separately for Scotland, England and Wales using country-relevant datasets and definitions of 'deep peat': soil with peat depth >40 cm in England and Wales and >50 cm in Scotland (SM 1.6).

Protected areas

Special Areas of Conservation (SACs) and Special Protection Areas (SPAs) represent high-quality conservation sites with enhanced protection under European legislation (JNCC, 2022a). Burning in PAs is contentious because site condition monitoring identifies burning as a contributor to poor ('unfavourable') condition of blanket bog in upland PAs (Natural England, 2008; NatureScot, 2023; Scottish Natural Heritage, 2010).

Spatial boundaries of PAs (JNCC, 2022b) were intersected with burn areas, and within each PA we calculated annual burned area and the percentage of available moorland burned ('available' defined as summed area of categories 'heather grassland', 'heather shrub' and 'bog' in Morton et al., 2021).

We also tested whether the percentage of available moorland burned differed inside and outside PAs. Because moorland inside and outside of PAs could differ in underlying environment (Douglas et al., 2015), we identified samples of land inside and outside PAs, matched by potentially confounding variables (peat

depth, slope, aspect, easting, northing, moorland extent) using the 'full matching' approach in the MatchIt R package (Ho et al., 2011) (SM 1.6). For this analysis, we summarised data at the scale of 500 × 500 m grid cells, a balance between dataset size and spatial resolution. The response variable was a proportion, created in R using cbind to join the number of burned and unburned pixels per grid cell (unburned = total number of moorland pixels per cell – number burned). This was modelled in a binomial Generalised Linear Model, with explanatory variables of PA status (inside/outside), year, their interaction, plus the covariates from the matching analysis ('doubly robust' analysis, Ramsey et al., 2019). This was conducted separately for Scotland and England due to the differences in legislation; relevant burn areas for Wales were insufficient. All covariates were standardised (mean zero and unit variance) before analysis. Results are presented as risk ratios; <1 indicates lower proportion of burning inside PAs than outside; 1 = no difference; >1 indicates a higher proportion of burning inside PAs.

Alpine habitats

Alpine habitats including moss heaths, dwarf-shrub heaths, grasslands and montane willow scrub are particularly sensitive to perturbations (Gordon et al., 2002; Tayside Biodiversity, 2015), and it is therefore important to assess their overlap with burning. UK alpine habitats generally occur above ~600 m elevation (Natural England, 2001; Tayside Biodiversity, 2015), but further North and West can be found at lower elevations (JNCC, 2023). To examine this overlap, we delineated alpine habitat using a bespoke vector dataset combining 'Alpine or sub-alpine' (EUNIS habitat classes E4 and F2) in the 20-m pixel Scotland Habitat and Land Cover Map 2020 (Scottish Government, 2021b) and the National Montane Bird Survey area (Hayhow et al., 2015; Whitfield, 2002) (see SM 1.6).

Slope

Guidance advises against moorland burning in sensitive areas, including any slopes >27°, or >18° in blanket mire or wet heath, due to the risk of erosion (Natural England, 2007; Scotland's Moorland Forum, 2017; Welsh Assembly Government, 2008). The slope of each burned pixel in our dataset was derived from OS terrain 50 data (Ordnance Survey, 2023) to evaluate compliance with this guidance. Where slope was >18° but <27°, overlap with habitat and peat depth data was used to identify relevant areas [i.e. in bog or heath LCM habitats (Morton et al., 2021) and over deep peat].

Results

Classification model evaluation

Models using spring indices of vegetation greenness (enhanced vegetation index [EVI], normalised difference vegetation index [NDVI], soil adjusted vegetation index [SAVI]) and burning (normalised burn ratio [NBR], mid-infrared burn index [MIRBI]) outperformed models using spring and summer or summer-only indices (SM 2.1). Our selected model (DNN2) had a balanced accuracy of 84.88%, while the overall accuracy was 96.71%. The FPR was 1.32%, FNR was 28.8% and MCC was 0.73. The performance of DNN1 and DNN2 was similar (SM 2.1), but due to a lower FPR and higher MCC, DNN2 was selected. There was little variation in accuracy metrics between years, although FNRs for 2020/21 and 2021/22 (mean 34.8%) were higher than other years (mean 22.7%) (SM 2.1; Tables S6 and S7). Burn extents were predicted using DNN2.

DNN2 predicted burn pixels in 28.1% of vegetation cut pixels (see Discussion).

Burning extent

Burning was mainly detected in Northern England, South-East Scotland and the Central and Eastern Scottish Highlands, with smaller areas in Wales (Figs. 2 and 3). Burn extents varied annually, with total annual burn areas across Great Britain between 8332 ± 1303 ha and $20\,973 \pm 1195$ ha, and an average of $15\,275 \pm 511$ ha year^{-1} (Table 1). Across all five of the burn seasons, the

greatest extent of burning took place in Scotland (59% of GB burned area [averaging 9047 ± 397 95% CI ha year^{-1}]), with 39% in England (averaging 5902 ± 104 ha year^{-1}) and 2% in Wales (325 ± 19 ha year^{-1}) (Table 1).

The total burned area showed a marked reduction in 2021/22, driven mainly by a reduction in area in England (Table 1; SM 2.2). The area burned in England in 2021/22 (1859 ± 240 ha) was 73% lower than the average of the four previous seasons (6913 ha) in England. This magnitude of reduction was not detected in Scotland or Wales (Table 1; SM 2.2).

Soil carbon

Across all five burn years, an average of 34% by area of burning in Great Britain occurred over deep peat, varying from 28% (5246 ± 778 ha) in 2020/21 to 41% (8552 ± 652 ha) in 2018/19 (Table 1). Within GB, 52% by area of burning over deep peat occurred in England (30–60% of the annual GB area between seasons), 47% in Scotland (40–70% between years) and 1% in Wales (1–4% between years). Although the absolute area of burning over deep peat in England declined in 2021/22, the percentage of burning over deep peat remained consistent with previous years (Table 1).

Protected areas

Overall, we detected >1 ha of burning in at least 1 year in 93 SACs and 43 SPAs. The North York Moors SPA/SAC and North Pennine Moors SPA/SAC were notable in terms

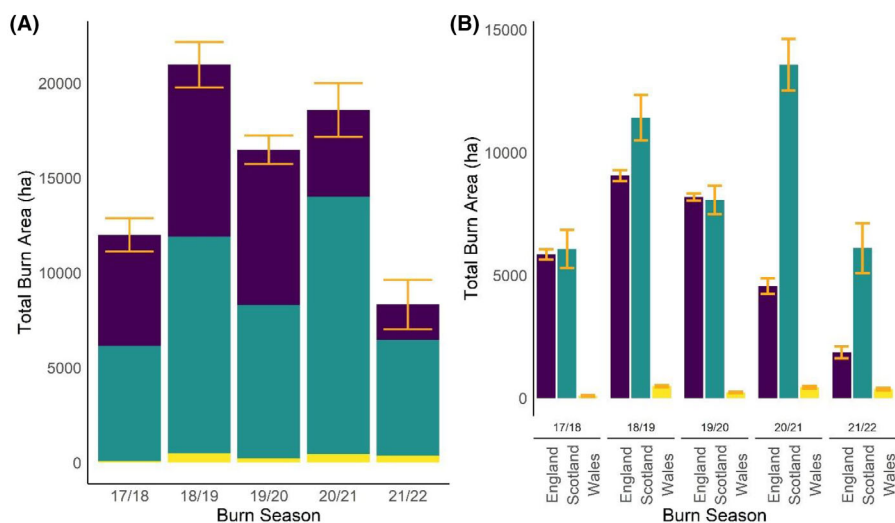


Figure 2. Area of moorland burning in Great Britain across five burn seasons; (A) total predicted burn area by country (blue = England, green = Scotland, yellow = Wales) (95% CI around total predicted area shown as red lines); (B) same data for each country separately to aid interpretation.

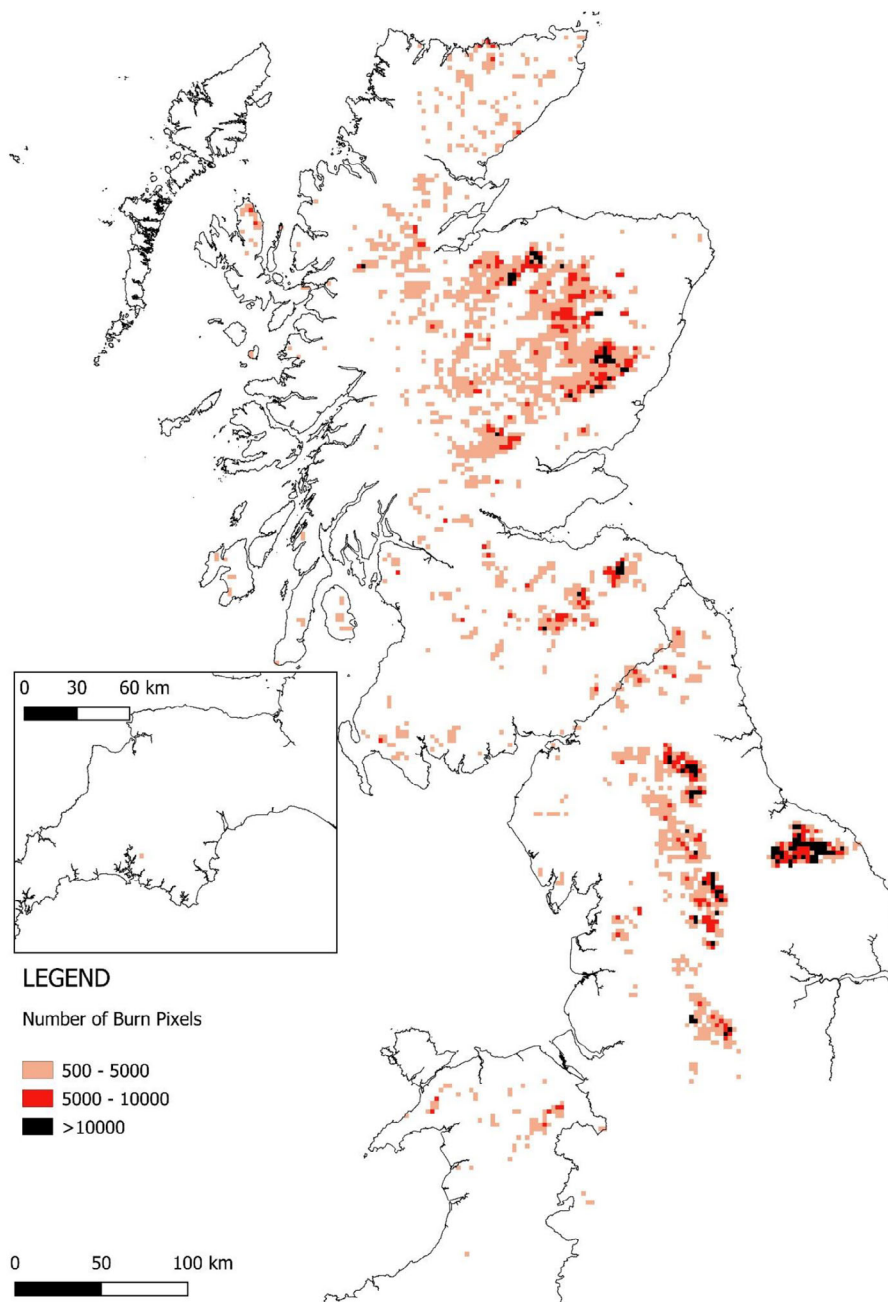


Figure 3. Extent of moorland burning across Great Britain in five (2017/18 to 2021/22) burn seasons. Due to the fine resolution at which burns are predicted (10-m pixels), burning is mapped here at 2.5 km grid square resolution (6.25 km^2) as the total number of pixels burned across the five seasons. Map shows only classified areas above northing 234 400 – the area containing most classified burn pixels with an inset map showing Dartmoor and Exmoor (the only other areas containing areas of classified burning). Grid squares with less than 5 ha of burning across the five seasons are transparent. Annual maps for the five burn seasons are at Figures S6–S10.

of area burned across the five burn years (Figs. 4 and 5). These sites, along with the Moorfoot Hills SAC, were also notable in terms of the percentage of moorland burned.

Over the five burn years, the area of burning in some PAs in England decreased (Figs. 4 and 5), with the

burned area in SPAs and SACs in 2021/22 66.7% and 65.2% lower, respectively, than the average of the four previous years. A similar pattern was observed in Scotland over the same period (SPAs: -41.5% , SACs: -36.6%), while in Wales burned area increased (8.6% in SACs,

Table 1. Classified annual burn extents and overlap with deep peat for Great Britain and constituent countries.

Year	GB (ha)				Scotland (ha)				England (ha)				Wales (ha)			
	Total	Deep peat (%)	Total (% of GB)	Mineral soil (%)	Shallow peat (%)	Deep peat (%)	Total (% of GB)	Mineral soil (%)	Shallow peat (%)	Deep peat (%)	Total (% of GB)	Mineral soil (%)	Shallow peat (%)	Deep peat (%)	Total (% of GB)	Mineral soil (%)
17/	12 001 ± 878	4772 ± 459 (40)	6072 ± 775 (51)	3933 ± 452 (65)	116 ± 19 (2)	2023 ± 304 (33)	5844 ± 209 (49)	2872 ± 88 (49)	239 ± 7 (4)	2733 ± 115 (47)	84 ± 34 (1)	2733 ± 115 (47)	239 ± 7 (4)	2733 ± 115 (47)	84 ± 34 (1)	2733 ± 115 (47)
18/	20 974 ± 1194	8552 ± 652 (41)	11 420 ± 929 (55)	7821 ± 449 (69)	187 ± 26 (2)	3412 ± 455 (30)	9061 ± 226 (43)	3610 ± 71 (40)	429 ± 10 (5)	5022 ± 145 (55)	493 ± 40 (2)	5022 ± 145 (55)	429 ± 10 (5)	5022 ± 145 (55)	493 ± 40 (2)	5022 ± 145 (55)
19/	16 487 ± 759	4834 ± 416 (29)	8071 ± 586 (49)	5931 ± 282 (74)	188 ± 17 (2)	1952 ± 147 (24)	8184 ± 147 (50)	5093 ± 47 (62)	211 ± 6 (3)	2880 ± 94 (35)	231 ± 27 (1)	2880 ± 94 (35)	211 ± 6 (3)	2880 ± 94 (35)	231 ± 27 (1)	2880 ± 94 (35)
20/	18 579 ± 1419	5246 ± 778 (28)	13 570 ± 1051 (73)	10 319 ± 511 (76)	145 ± 30 (1)	3106 ± 511 (23)	4562 ± 313 (25)	2250 ± 99 (49)	250 ± 14 (6)	2062 ± 200 (45)	446 ± 54 (2)	2062 ± 200 (45)	250 ± 14 (6)	2062 ± 200 (45)	446 ± 54 (2)	2062 ± 200 (45)
21/	8333 ± 1303	2586 ± 389 (31)	6103 ± 1016 (73)	4195 ± 487 (69)	108 ± 29 (1)	1800 ± 500 (30)	1859 ± 240 (22)	1018 ± 56 (55)	75 ± 6 (4)	766 ± 179 (41)	371 ± 47 (4)	766 ± 179 (41)	75 ± 6 (4)	766 ± 179 (41)	371 ± 47 (4)	766 ± 179 (41)
22/	15 275 ± 510	5198 ± 250 (34)	9047 ± 397 (59)	6440 ± 198 (71)	149 ± 11 (1)	2459 ± 189 (27)	5902 ± 104 (39)	2969 ± 33 (50)	24 ± 4 (4)	2693 ± 68 (46)	325 ± 19 (2)	2693 ± 68 (46)	24 ± 4 (4)	2693 ± 68 (46)	325 ± 19 (2)	2693 ± 68 (46)
Mean																

Deep peat defined as Scotland >50 cm; England and Wales >40 cm in depth; shallow peat (0 – country-specific deep peat threshold) and mineral soil (0 cm peat) also shown. Burn extents are rounded to the nearest ha. 95% CI are shown (±) with percentages of country totals shown for each peat depth category and as percentages of the GB total burn area for country totals in brackets.

37.5% in SPAs) although the total burn areas in Welsh PAs were small (<130 ha).

The relative proportion of available moorland burned inside and outside PAs differed between years for both England and Scotland (Table 2). In England, a significantly greater proportion of available moorland was burned inside PAs than outside in 2 of 5 years; in 1 year, a significantly greater proportion of available moorland was burned outside PAs than inside, and in 2 years there was no significant difference inside and outside PAs (Table 2a; Fig. 6). In Scotland, the proportion of moorland burned inside PAs was significantly lower than that outside PAs in all years (Table 2b).

Alpine habitats

Some 158 ha of burning (0.2% of total burned area) occurred in alpine habitats over the 5 years (Table 3), and this varied little between years (26–37 ha, mean of 32 ha [SM 2.5]). Alpine burning occurred mainly in the Cairngorms and Northwest Highlands of Scotland, with smaller areas in the Lammermuir Hills of South-east Scotland (Fig. 7). The 100 m elevation band in which the greatest area of burning in alpine habitats occurred was 650–750 m, although burning in alpine habitats, as identified by EUNIS, was detected from 250–350 m up to 950–1050 m bands (Table 3).

Slope

Some 915 ha of burning occurred on steep slopes >27°, with 258 ha on slopes >18° in blanket mire or wet heath (1.6% of total burn extent) over the five burn seasons. Burning on steep slopes occurred mainly in the Cairngorms, with additional areas including the North York Moors, Lammermuir Hills and Northwest Highlands (Fig. 8). The annual percentage of burning on slopes >27° or >18° over deep peat showed no obvious trend over the 5 years (SM 2.6).

Discussion

Our moorland burning detection method has high accuracy (~97%, MCC = 0.73). Performance was better using imagery from the spring period immediately following the end of the burn season than imagery from the summer. This is likely due to vegetation regrowth within burn scars, with increasing time since burning reducing ability to separate the spectral signature in burned and unburned areas.

Our method could not fully separate burning from cutting on moorland, with 28.1% of a sample of known cuts mapped as burns. This suggests that while the reduction

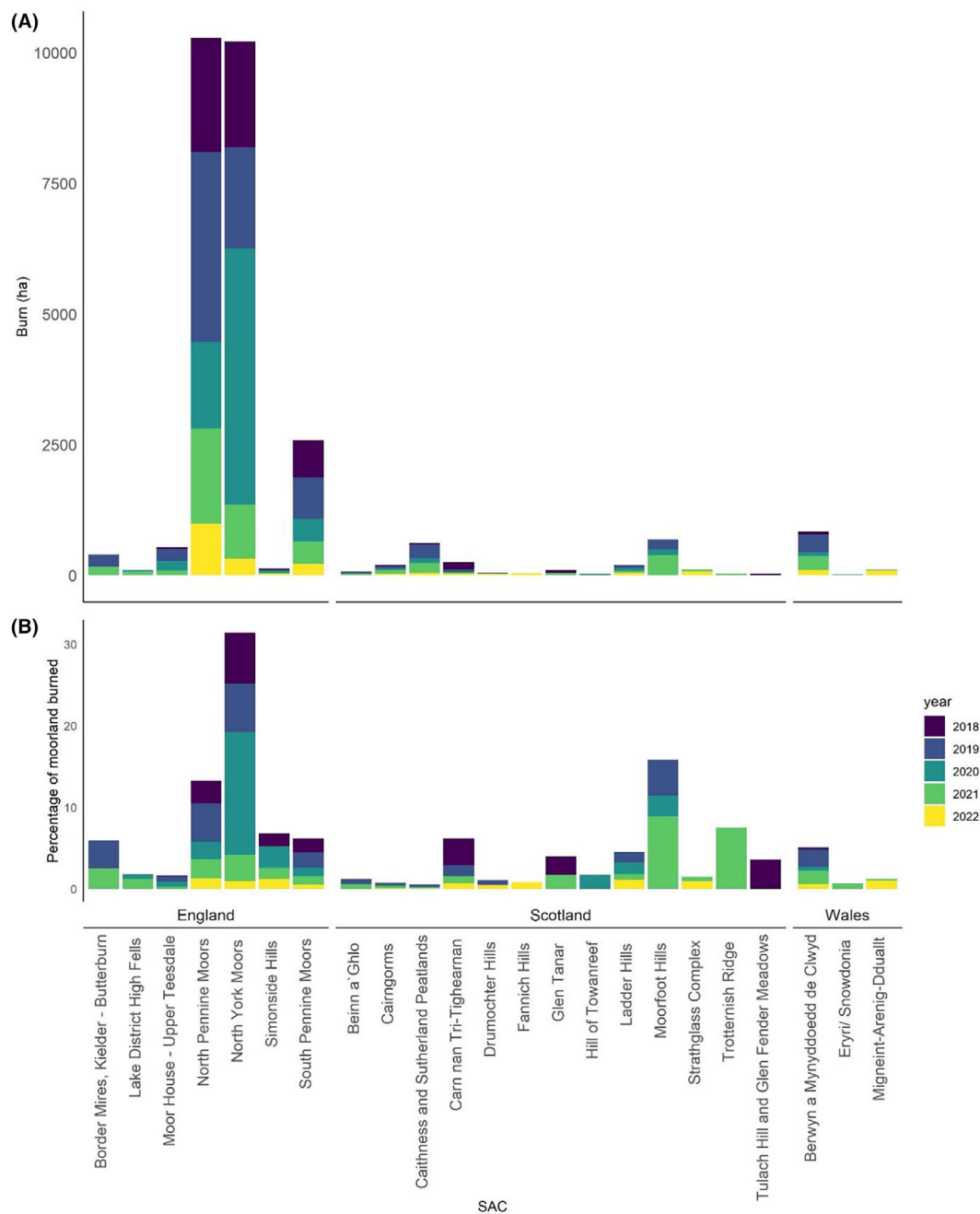


Figure 4. Burning extent in special areas of conservation (SACs) designated for their habitat conservation value. Shown are SACs with >20 ha burned in at least 1 year: (A) total burn area; (B) percentage area of moorland burned.

in photosynthesising vegetation following burning or cutting is mostly distinguishable, the spectral signature in some cuts and burns is similar. It is worth noting, however, that while there are anecdotal reports of increasing use of cutting (GWCT, 2023b), as burning is reduced by legislative change, we still expect most vegetation scars detected in our study to comprise burning. While the

inclusion of some cuts in the total area burned could inflate the area burned, it is important to note that this is likely to be 28% of a relatively small area of cutting across moorland in Great Britain. Furthermore, inclusion of some cuts is likely to be more than offset by our conservative approach to model selection, where we minimised false positives and accepted higher false negatives (missed

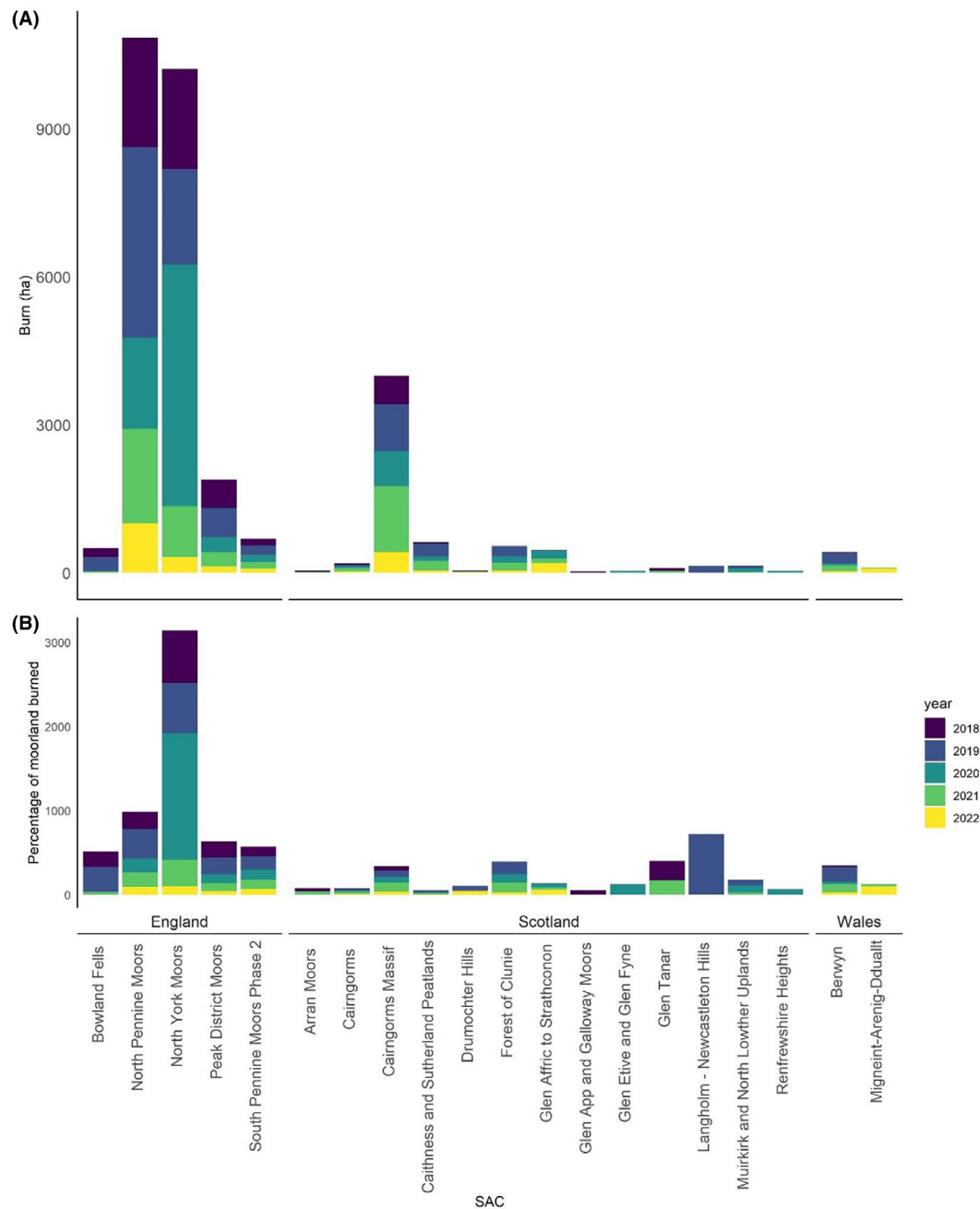


Figure 5. Burning extent in special protection areas (SPAs) designated for their ornithological conservation value. Shown are SPAs with >20 ha burned in at least 1 year: (A) total burn area; (B) percentage area of moorland burned.

burns). Consequently, cutting would need to be extensive and not displace significant areas of existing burning to have a significant impact on classified total burn extents. We therefore do not consider the inclusion of a proportion of cuts to have markedly influenced overall totals of burn extent, and our estimates will still be lower than true annual burn extents. Refinement of our method by

validation with larger samples of known burns and cuts could improve the automated classification, and this could be further enhanced through the use of other data sources (e.g. landownership, aerial imagery) or image analysis methods (e.g. Convolutional Neural Networks for identifying the tracks from cutting vehicles) to enhance separation.

Table 2. Moorland burning inside and outside of protected areas (PAs).

Country	Contrast	Burn season	Risk ratio	CI 2.50%	CI 97.50%	<i>P</i> (> z)
(a)						
England	In PA/Out of PA	2017/18	1.05	0.682	1.616	0.826
England	In PA/Out of PA	2018/19	0.621	0.483	0.798	<0.001
England	In PA/Out of PA	2019/20	1.925	1.555	2.383	<0.001
England	In PA/Out of PA	2020/21	1.364	1.1	1.691	0.0045
England	In PA/Out of PA	2021/22	1.266	0.945	1.695	0.114
(b)						
Scotland	In PA/Out of PA	2017/18	0.495	0.426	0.575	<0.001
Scotland	In PA/Out of PA	2018/19	0.647	0.578	0.724	<0.001
Scotland	In PA/Out of PA	2019/20	0.62	0.567	0.679	<0.001
Scotland	In PA/Out of PA	2020/21	0.692	0.646	0.742	<0.001
Scotland	In PA/Out of PA	2021/22	0.511	0.459	0.569	<0.001

Comparisons are made on a sample of 500 m grid cells (England: $n = 3889$, Scotland: $n = 52\ 133$) matched by potentially confounding environmental covariates. Risk ratio of 1 indicates comparable extents of burning (relative to available moorland) inside and outside PAs; >1 indicates proportionally more burning inside PAs and <1 proportionally more burning outside PAs. Where confidence intervals (CI) overlap 1, there is no statistically significant difference in proportional area of moorland burned inside and outside PAs.

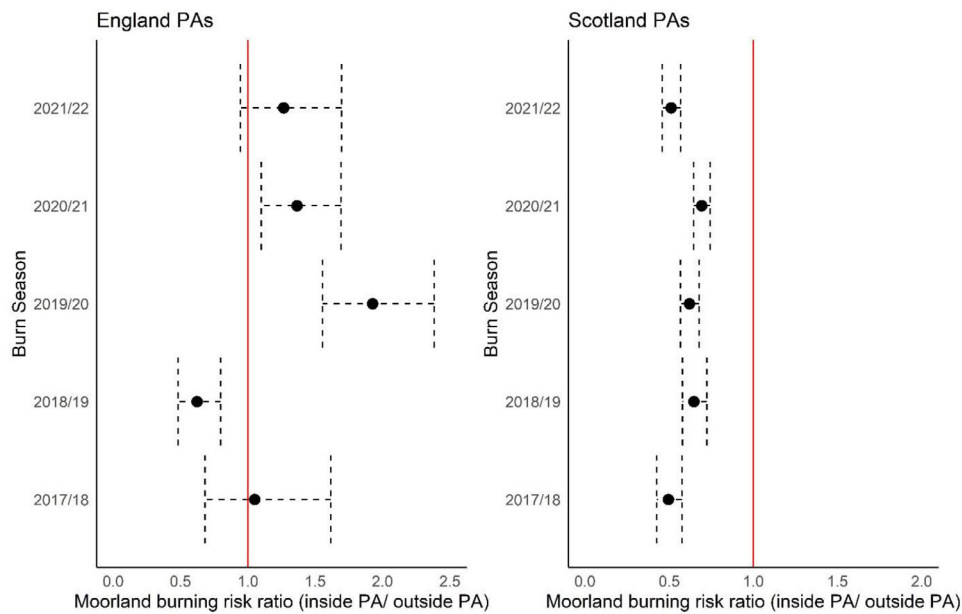


Figure 6. Comparison of moorland burning inside and outside protected areas (PAs) relative to the area of moorland available in England and Scotland. Comparisons are made on a sample of 500 m grid cells. A risk ratio value of 1 (vertical red line) indicates comparable proportional extents of burning inside and outside PAs, values >1 indicate a greater proportion of burning inside PAs, values <1 indicate lower proportion inside PAs. Black dots ($\pm 95\%$ CI) show mean across PAs per year.

Our method did not separate burning for Red Grouse shooting from burning for grazing management. However, most burning undertaken in the UK uplands is considered to be for Red Grouse shooting (Yallop et al., 2006), and upland burn maps are used as indices of grouse moor management (e.g. Anderson et al., 2009; Ewing et al., 2023; Franks et al., 2017; Murgatroyd et al., 2019). The model did not separate prescribed

burning from wildfires. However, as an example, the reported annual average area of upland wildfires in English mountain, heath and bog habitat (2009–10 to 2020–21) is 89 ha year^{-1} (Forestry Commission, 2023) and thus would account for no more than 1–1.5% of our English-mapped burned area. Furthermore, ~25% of upland wildfires are considered to originate from escaped prescribed burns (Worrall et al., 2010), and in cases

Table 3. (a) Classified burn extents (ha) in alpine habitat, slopes $>18^\circ$ on wet heath or bog and slopes $>27^\circ$ by year; (b) classified burn extents (ha) per 100 m elevation band across all burn seasons 2017/18 to 2021/22.

Burn season	Classified burned area (ha) in alpine habitats	Classified burned area (ha) on slopes >18° on wet heath or bog habitat	Classified burned area (ha) on slopes >27°
(a)			
2017/18	37	25	87
2018/19	31	56	104
2019/20	35	40	194
2020/21	26	95	319
2021/22	29	43	211
Elevation (m)		Total classified burn area (ha)	
(b)			
250–350		9.58	
350–450		14.23	
450–550		18.75	
550–650		29.9	
650–750		44.85	
750–850		36.33	
850–950		2.38	
950–1050		0.05	

where the cause of ignition was recorded this increased to 68% (Glaves et al., 2020), escaped prescribed burns should therefore be included in metrics of annual prescribed burn extent.

Spatial patterns in burning

Our estimate of the annual area burned in England (average 5902 ha, range 1859–9061 ha) is lower than a previous estimate of 114 km² (~11 400 ha) from 2000 (Yallop et al., 2006). The latter was, however, extrapolated from a 2% sample of the English uplands to a national estimate, which could produce variability around the estimate. Our results are the first systematic estimates of annual area burned across the uplands of Great Britain and constituent countries, and across multiple consecutive years. It is important to note that burning for Red Grouse shooting takes place on ~8–25-year rotations (Thacker et al., 2015). Therefore, our data from 5 years will likely underestimate (i) the total area of the uplands under rotational burning in recent decades, (ii) the total area therefore managed for Red Grouse shooting, and (iii) the total areas of

moorland over deep peat, in PAs, in alpine habitats and on steeply sloping ground under burning rotations.

Temporal patterns in burning

The 73% reduction in burned area in England in 2021/22 relative to the four previous annual burn seasons is notable. This reduction was consistent in magnitude whether using the burned area predicted from any pixels useable in a given season or only those useable in all five seasons (i.e. not obscured by cloud or snow in any year), suggesting it is a real change (SM 2.2). Although the area burned in Scotland was relatively low in 2021/22, it was not the lowest annual Scottish extent of burning recorded across the five seasons, suggesting that the reduction is largely England-specific. This reduction coincides with the introduction of regulation in England on 1 May 2021 to limit burning over deep peat within protected sites (UK Government, 2021), suggesting it may have contributed to the reduced extent of burning there.

The relatively low burned area in England in 2021/22 could also be due to other factors: (i) weather, including rainfall which could limit burning activity, although summary rainfall data for the five burn seasons suggest that 2021/22 was not markedly wetter (Hollis et al., 2018; SM 3), and (ii) changes in burning practice (e.g. smaller burns) that could affect their detectability with 10 m resolution imagery. However, recommended burn sizes remain 2–5 ha (DEFRA, 2007; Scottish Government, 2011; Welsh Government, 2008), and we consider a widespread shift to smaller burns, in one season, unlikely to have driven the observed patterns; (iii) a reduction in the temporal transferability of our model, as we move further away from the years of training data (2018/19 and 2019/20). This could yield progressive increases in the false negative rate, which potentially explains the observed temporal pattern (SM 2.1). However, any resulting reduction in burn detection should be widely spatially consistent, not just in England. We therefore consider it unlikely that the observed patterns originate from a lack of temporal transferability. Nevertheless, future iterations of our approach could test the temporal transferability of models with multi-year trained models (Tuanmu et al., 2011), the direct incorporation of phenological variables (e.g. Gao et al., 2022) and finer-resolution imagery (e.g. 3 m PlanetScope, Planet Labs PBC, 2023).

Burning and soil carbon

The percentage of burning over deep peat across Great Britain, England and Scotland, respectively (34, 46, 27%), is similar to previous estimates for England and Scotland combined and separately (33.5, 46.3, 27.6%, respectively;

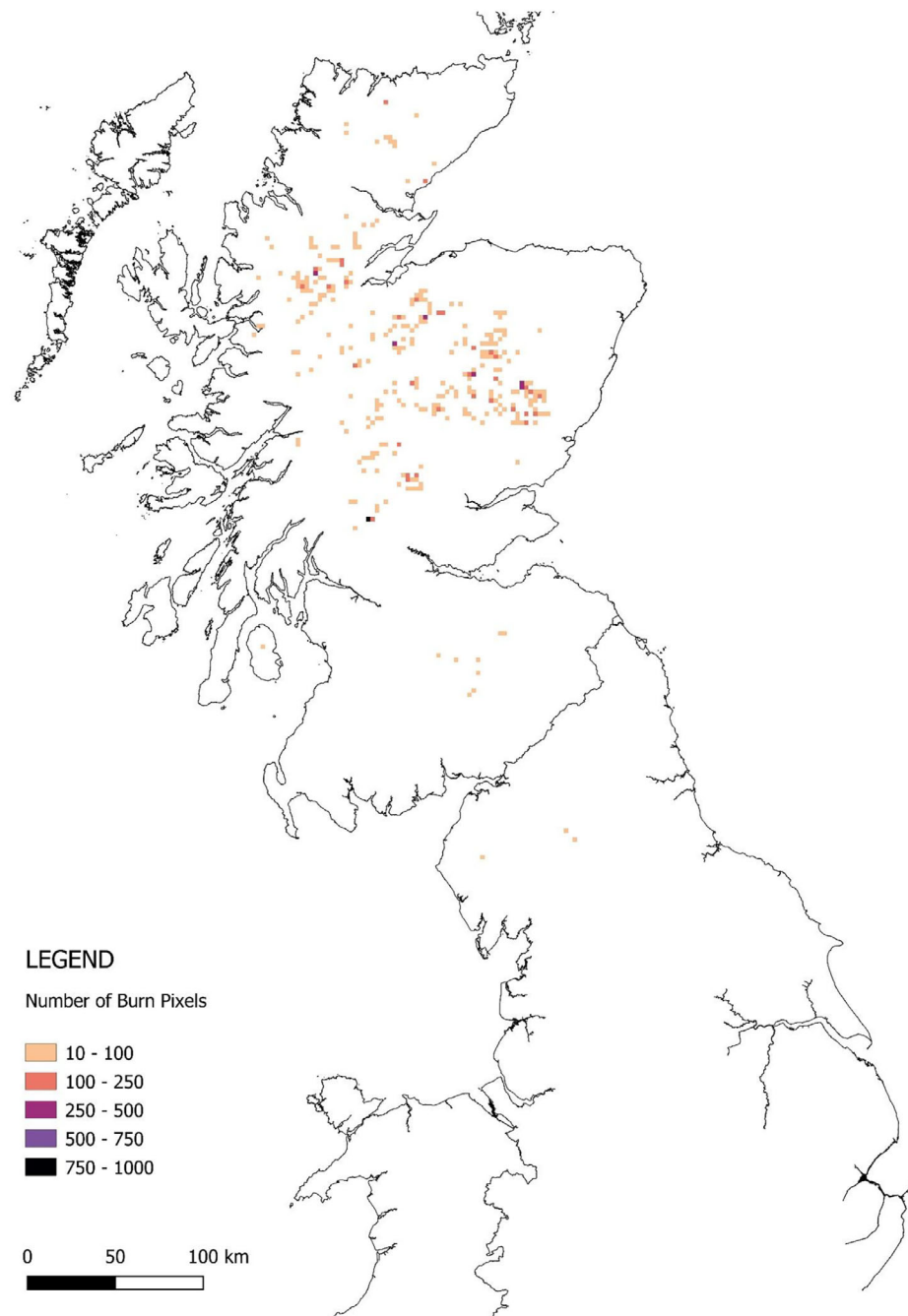


Figure 7. Extent of burning within alpine habitats in Great Britain. Due to the fine resolution at which burns are predicted (10-m pixels), burning is mapped at the resolution of 2.5 km grid squares (6.25 km²) as the total number of pixels burned across the five burn seasons.

Douglas et al., 2015, also 34% in Scotland from 1985 to 2022; Spracklen & Spracklen, 2023). Although the absolute area of burning over deep peat reduced in England over our five burn seasons, it reduced by a relatively small amount as a percentage of all burning (to 41% in 2021/22). Thus, a notable proportion of burning in England still takes place over deep peat. This could be due to

non-compliance with the new (UK Government, 2021) legislation inside PAs (e.g. Crown Prosecution Service, 2023) and/or because deep peat outside PAs in England is not protected by the 2021 legislation. In Scotland, the percentage of all burning over deep peat showed no notable change over the 5 years (consistent with a longer-term study; Spracklen & Spracklen, 2023). This

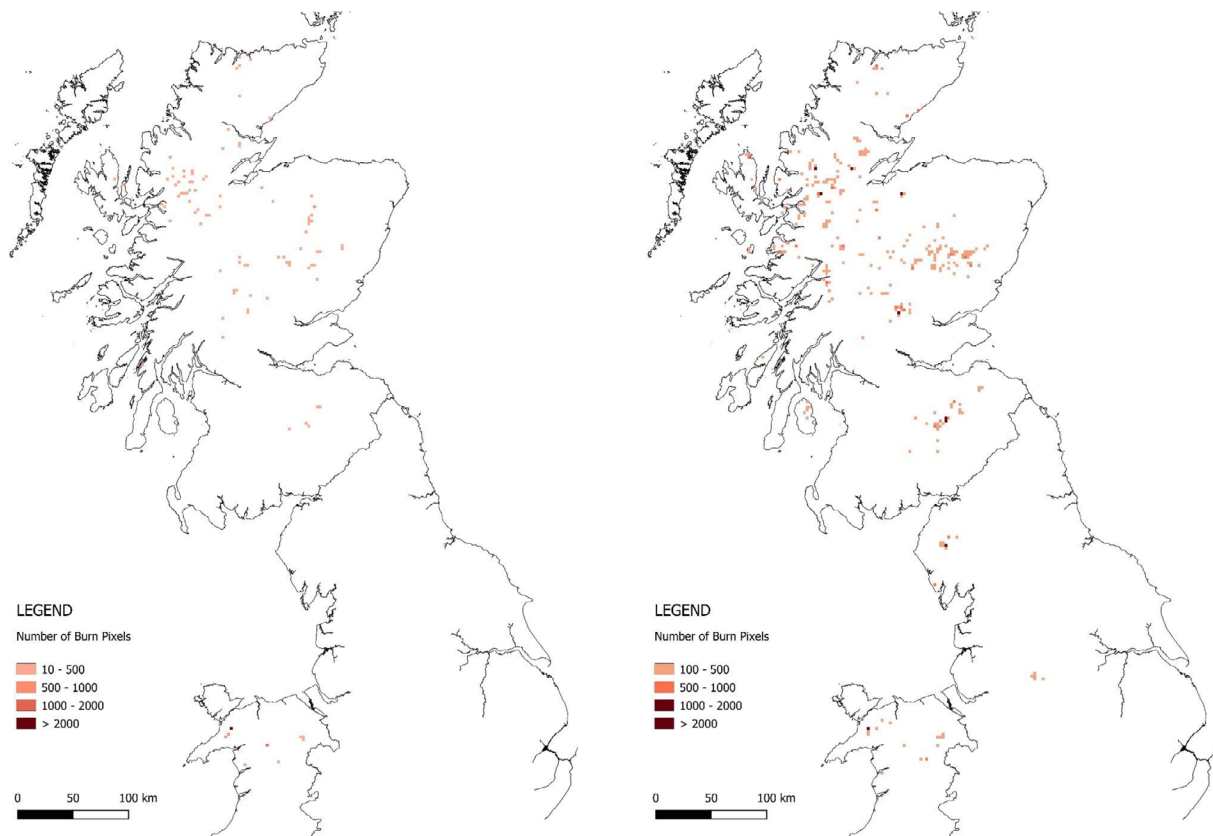


Figure 8. Number of burn pixels per 2.5 km grid square on steep slopes for (A) slopes greater than 18° (where over deep peat and on moorland habitats) and (B) slopes greater than 27°, across five burn seasons between 2017/18 and 2021/22. Due to the fine resolution at which burns are predicted (10-m pixels), burning is mapped as the total number of pixels burned across the five burn seasons per 2.5-km grid square.

suggests that a voluntary reduction in burning over peat soils is unlikely and that upcoming legislation to licence burning on peatlands (Scottish Parliament, 2023) is required to improve the sustainability of moorland management.

Protected areas

Protected areas in the North of England were subject to the highest percentage area of burning, a result consistent with Douglas et al. (2015). The most intensively burned sites (defined as percentage area of moorland where burning was detected) in Douglas et al. (2015) were the North Pennine Moors SAC and SPA (11.5 and 10.8%, respectively). While our results are not directly comparable with these (due to methodological differences), we detected similar percentages for total moorland area burned in these sites (SAC 13.3%, SPA 9.8%) over the five burn seasons. Unlike Douglas et al. (2015), we found that the North York Moors SAC/SPA (SAC 31.4%, SPA 31.4%) were the most intensively burned (by area) PAs. Whether this reflects a recent intensification in burning area in the

North York Moors, or is due to the different burn detection methods, is unclear.

The area burned in protected sites reduced in both England and Scotland in the 2021/22 burn season compared to the four previous years (SPAs –66.7% and 41.5%, SACs –65.2% and 36.6%, respectively). The greater reduction in England is notable and may be due to the introduction of legislation limiting burning over deep peat in PAs by the UK Government (2021). If true, it suggests that regulation of moorland burning may have been an effective policy in England, at least in PAs, and could be effective if introduced elsewhere in the UK. However, it would be necessary to repeat our mapping exercise in future years to test whether the reduction in burning in England is sustained rather than a one-off event.

The proportion of moorland burned inside and outside PAs showed contrasting patterns between countries. In Scotland, a consistently lower proportion of available moorland was burned inside PAs than outside. In contrast, in England, a higher proportion of moorland was burned inside PAs in two out of 5 years. In 2021/22, following regulation to limit burning over deep peat in

English PAs, it is noteworthy that there was no significant difference in the proportion of available moorland burned inside and outside of PAs, when a reduction in PAs might have been expected. Douglas et al. (2015) previously found a higher proportion of moorland burned inside PAs but did not analyse differences between countries or years.

Protected area site condition monitoring identifies inappropriate burning as a significant contributor to poor ('unfavourable') condition of blanket bog in upland PAs in England (Natural England, 2008) and Scotland (NatureScot, 2023; Scottish Natural Heritage, 2010). Our results re-confirm the continued presence of burning across numerous sites in the UK's PA network and highlight the need for ongoing site condition monitoring. Continued monitoring of burning patterns for this reason alone may be essential.

Location of burns and guidance compliance

The area of burning in alpine habitats was low relative to that of other moorland types and varied relatively little between years (26–37 ha annually). It is important to continue to monitor spatial and temporal patterns of alpine burning, because these habitats are considered particularly sensitive to disturbance (Gordon et al., 2002) and face threats including climate change (Watts et al., 2022), atmospheric pollution (Pearce & van der Wal, 2002) and inappropriate grazing (Natural England, 2001). We are not aware of current guidance on burning in alpine habitats, although a government advisory body previously stated 'There are no benefits from burning in the montane zone, for biodiversity or for agricultural or game management' (Natural England, 2001).

While the area of burning on sloping ground $>27^\circ$ (or $>18^\circ$ on 'deep peat') represents a relatively small percentage (1.6%) of total burn extent, it contravenes moorland burning guidance (Natural England, 2007; Scotland's Moorland Forum, 2017; Welsh Assembly Government, 2008). Furthermore, burning on steep slopes could have more pronounced environmental impacts due to the potential for increased fire severity (Scotland's Moorland Forum, 2017; Scottish Government, 2013), higher post-fire soil erosion (DeBano, 2000) and reduced rainfall infiltration (Doerr et al., 2009). It has also been suggested that post-fire vegetation recovery is inhibited on steeper slopes (Meneses, 2021).

The Muirburn Code (Scotland's Moorland Forum, 2017) also recommends a fire-free buffer zone of 2–5 m wide for watercourses and lochs (depending on their size). Unfortunately, due to the spatial resolution of our classification, quantifying compliance was not possible. Such an analysis would be informative but would require higher resolution

imagery, perhaps best focused within a sub-sample of areas of moorland managed through burning.

Conclusions and policy implications

Our work provides early evidence that the regulation of burning in England may be having the desired effect of reducing burned area and that the previously sought voluntary agreements were ineffective. It should be noted, however, that regulations in England apply only to burning over deep peat in PAs, leaving 4% of upland deep peat unprotected. Given the proposed regulation of burning on peatland in Scotland, so that it is only permitted in very limited circumstances (Scottish Parliament, 2023), our method could be used to assess changes in burning in Scotland. The continued overlap of burning with peat soils and steeper slopes suggests that burning in the UK uplands continues to fall short of recommended sustainable practices. Our method will enable regular re-assessment of burning extents and overlap with ecosystem services.

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References

- Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C. et al. (2016) Tensorflow: large-scale machine learning on heterogeneous distributed systems. *arXiv preprint arXiv:1603.04467*.
- Allen, K.A., Denelle, P., Ruiz, F.M.S., Santana, V.M. & Marrs, R.H. (2016) Prescribed moorland burning meets good practice guidelines: a monitoring case study using aerial photography in the Peak District, UK. *Ecological Indicators*, **62**, 76–85.
- Anderson, B.J., Arroyo, B.E., Collingham, Y.C., Etheridge, B., Fernandez-De-Simon, J., Gillings, S. et al. (2009) Using distribution models to test alternative hypotheses about a species' environmental limits and recovery prospects. *Biological Conservation*, **142**(3), 488–499.
- Bain, C.G., Bonn, A., Stoneman, R., Chapman, S., Coupar, A., Evans, M. et al. (2011) *IUCN UK commission of inquiry on peatlands*. Edinburgh IUCN UK Peatland Programme.
- Boughorbel, S., Jarray, F. & El-Anbari, M. (2017) Optimal classifier for imbalanced data using Matthews correlation coefficient metric. *PLoS One*, **12**(6), e0177678.
- Brown, L.E., Holden, J., Palmer, S.M., Johnston, K., Ramchunder, S.J. & Grayson, R. (2015) Effects of fire on the

- hydrology, biogeochemistry, and ecology of peatland river systems. *Freshwater Science*, **34**(4), 1406–1425.
- Brownlee, J. (2020) *Imbalanced classification with python: better metrics, balance skewed classes, cost-sensitive learning*. Machine Learning Mastery.
- Climate Change Committee. (2020) *Land use: policies for a net zero UK*. Available from: <https://www.theccc.org.uk/publication/land-use-policies-for-a-net-zero-uk/> [Accessed 8th August 2023].
- Crown Prosecution Service. (2023) *Landowning company fined in ground-breaking case*. Available from: <https://www.cps.gov.uk/mersey-cheshire/news/landowning-company-fined-ground-breaking-case> [Accessed 3rd April 2024].
- DeBano, L.F. (2000) The role of fire and soil heating on water repellency in wildland environments: a review. *Journal of Hydrology*, **231**, 195–206.
- DEFRA. (2007) *The heather and grass burning code*. SI 2021/158.
- DEFRA. (2021) *England peat action plan*. Available from: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1010786/england-peat-action-plan.pdf [Accessed 8th August 2023].
- Doerr, S.H., Shakesby, R.A. & MacDonald, L.H. (2009) Soil water repellency: a key factor in post-fire erosion. In: Cerda, A. & Robichaud, P.R. (Eds.) *Fire effects on soils and restoration strategies*. Boca Raton: CRC Press, pp. 213–240.
- Douglas, D.J., Beresford, A., Selvidge, J., Garnett, S., Buchanan, G.M., Gullett, P. et al. (2017) Changes in upland bird abundances show associations with moorland management. *Bird Study*, **64**(2), 242–254.
- Douglas, D.J., Buchanan, G.M., Thompson, P., Amar, A., Fielding, D.A., Redpath, S.M. et al. (2015) Vegetation burning for game management in the UK uplands is increasing and overlaps spatially with soil carbon and protected areas. *Biological Conservation*, **191**, 243–250.
- Ewing, S.R., Thomas, C.E., Butcher, N., Denman, B., Douglas, D.J., Anderson, D.I. et al. (2023) Illegal killing associated with gamebird management accounts for up to three-quarters of annual mortality in hen harriers *Circus cyaneus*. *Biological Conservation*, **283**, 110072.
- Forestry Commission. (2023) *Forestry Commission Wildfire statistics for England: report to 2020–21 – Summary*. Available from: <https://www.gov.uk/government/publications/forestry-commission-wildfire-statistics-for-england-report-to-2020-21/forestry-commission-wildfire-statistics-for-england-report-to-2020-21-summary#:~:text=Fire%20and%20Rescue%20Services%20attended,over%206%2C600%20hectares%20per%20annum> [Accessed 9th August 2023].
- Franks, S.E., Douglas, D.J., Gillings, S. & Pearce-Higgins, J.W. (2017) Environmental correlates of breeding abundance and population change of Eurasian curlew *Numenius arquata* in Britain. *Bird Study*, **64**(3), 393–409.
- Gao, J., Holden, J. & Kirkby, M. (2017) Modelling impacts of agricultural practice on flood peaks in upland catchments: an application of the distributed TOPMODEL. *Hydrological Processes*, **31**(23), 4206–4216.
- Gao, Z., Guo, D., Ryu, D. & Western, A.W. (2022) Enhancing the accuracy and temporal transferability of irrigated cropping field classification using optical remote sensing imagery. *Remote Sensing*, **14**(4), 997.
- Glaves, D.J., Crowle, A.J., Bruemmer, C. & Lenaghan, S.A. (2020) *The causes and prevention of wildfire on heathlands and peatlands in England*. Report number: Natural England Evidence Review NEER014.
- Glaves, D.J., Morecroft, M., Fitzgibbon, C., Leppitt, P., Owen, M. & Phillips, S. (2013) *The effects of managed burning on upland peatland biodiversity, carbon and water*. Natural England Evidence Review 004. Natural England, Sheffield.
- Gordon, J.E., Dvůrák, I.J., Jonasson, C., Josefsson, M., Kociánová, M. & Thompson, D.B. (2002) Geo-ecology and management of sensitive montane landscapes. *Geografiska Annaler: Series A, Physical Geography*, **84**(3–4), 193–203.
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D. & Moore, R. (2017) Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, **202**, 18–27.
- GWCT. (2023a) *Heather burning*. Available from: <https://www.gwct.org.uk/policy/briefings/driven-grouse-shooting/heather-burning/> [Accessed 8th August 2023].
- GWCT. (2023b) *What effect does heather cutting have on mosses?*
- Harper, A.R., Doerr, S.H., Santin, C., Froyd, C.A. & Sinnadurai, P. (2018) Prescribed fire and its impacts on ecosystem services in the UK. *Science of the Total Environment*, **624**, 691–703.
- Hayhow, D.B., Ewing, S.R., Baxter, A., Douse, A., Stanbury, A., Whitfield, D.P. et al. (2015) Changes in the abundance and distribution of a montane specialist bird, the dotterel *Charadrius morinellus*, in the UK over 25 years. *Bird Study*, **62**(4), 443–456.
- Heinemeyer, A., Berry, R. & Sloan, T.J. (2019) Assessing soil compaction and micro-topography impacts of alternative heather cutting as compared to burning as part of grouse moor management on blanket bog. *PeerJ*, **7**, e7298. Available from: <https://doi.org/10.7717/peerj.7298>
- Hijmans, R.J., Bivand, R., Forner, K., Ooms, J., Pebesma, E. & Sumner, M.D. (2022) *Package 'terra'*. Vienna, Austria: Maintainer.
- Ho, D., Imai, K., King, G. & Stuart, E.A. (2011) MatchIt: nonparametric preprocessing for parametric causal inference. *Journal of Statistical Software*, **42**(8), 1–28. Available from: <https://doi.org/10.18637/jss.v042.i08>
- Holden, J., Shotbolt, L., Bonn, A., Burt, T.P., Chapman, P.J., Dougill, A.J. et al. (2007) Environmental change in moorland landscapes. *Earth-Science Reviews*, **82**(1–2), 75–100.
- Holland, J.P., Pollock, M., Buckingham, S., Glendinning, J. & McCracken, D. (2022) *Reviewing, assessing and critiquing the*

- evidence base on the impacts of burn on wildfire prevention, carbon storage and biodiversity. NatureScot Research Report No. 1302.
- Hollis, D., McCarthy, M., Kendon, M., Legg, T. & Simpson, I. (2018) *HadUK-Grid gridded and regional average climate observations for the UK*. Centre for Environmental Data Analysis. Available from: <http://catalogue.ceda.ac.uk/uuid/4dc8450d889a491ebb20e724debe2dfb> [Accessed 3rd April 2024].
- IUCN UK Peatland Programme. (2023) *Position statement: burning and Peatlands*. Available from: https://www.iucn-uk-peatlandprogramme.org/sites/default/files/2023-04/Position%20Statement%20-%20Burning%20and%20Peatlands%20V4%20-%20FINAL_1.pdf [Accessed 8th August 2023].
- JNCC. (2022a) *UK protected areas*. Available from: <https://jncc.gov.uk/our-work/uk-protected-areas/> [Accessed 3rd April 2024].
- JNCC. (2022b) *UK protected area datasets for download*. Available from: <https://jncc.gov.uk/our-work/uk-protected-area-datasets-for-download/> [Accessed 19th May 2023].
- JNCC. (2023) *4060 Alpine and boreal heaths*. Available from: <https://sac.jncc.gov.uk/habitat/H4060/> [Accessed 30th August 2023].
- Lucas, R., Medcalf, K., Brown, A., Bunting, P., Breyer, J., Clewley, D. et al. (2011) Updating the phase 1 habitat map of Wales, UK, using satellite sensor data. *ISPRS Journal of Photogrammetry and Remote Sensing*, **66**(1), 81–102.
- Matthews, K.B., Fielding, D., Miller, D.G., Gandossi, G., Newey, S. & Thomson, S. (2020) *Mapping the areas of moorland that are actively managed for grouse and the intensity of current management regimes. Part 3 – Research to assess socioeconomic and biodiversity impacts of driven grouse moors and to understand the rights of gamekeepers*. Commissioned report for Scottish Government, p. 47. Available from: <https://sefari.scot/sites/default/files/documents/Part%203%20-%20GIS%20Mapping%20of%20Grousemoor.pdf> [Accessed 3rd April 2024].
- Meneses, B.M. (2021) Vegetation recovery patterns in burned areas assessed with landsat 8 OLI imagery and environmental biophysical data. *Fire*, **4**(4), 76.
- Morton, R.D., Marston, C.G., O'Neil, A.W. & Rowland, C.S. (2021) *Land cover map 2020 (10 m classified pixels, GB)*. NERC EDS Environmental Information Data Centre [Data set]. Available from: <https://doi.org/10.5285/35c7d0e5-1121-4381-9940-75f7673c98f7>
- Murgatroyd, M., Redpath, S.M., Murphy, S.G., Douglas, D.J., Saunders, R. & Amar, A. (2019) Patterns of satellite tagged hen harrier disappearances suggest widespread illegal killing on British grouse moors. *Nature Communications*, **10**(1), 1094.
- Murphy, K.A., Reynolds, J.H. & Koltun, J.M. (2008) Evaluating the ability of the differenced normalized burn ratio (dNBR) to predict ecologically significant burn severity in Alaskan boreal forests. *International Journal of Wildland Fire*, **17**(4), 490–499.
- Natural England. (2001) *The upland management handbook (SC26)*. Available from: <https://publications.naturalengland.org.uk/publication/82050> [Accessed 3rd April 2024].
- Natural England. (2007) *The heather and grass burning code*. London: Natural England. Available from: <https://gfmco.uk/online/programmes/natcon/UK-DEFRA--Heather-Grass-Burning-Code-2007.pdf> [Accessed 28th April 2023].
- Natural England. (2008) *State of the Natural Environment 2008 (NE85)*. Available from: <https://webarchive.nationalarchives.gov.uk/ukgwa/20140304124520/http://publications.naturalengland.org.uk/publication/31043?category=118044> [Accessed 27th June 2023].
- Nature Scot. (2023) *The proportion of Scotland's protected sites in favourable condition 2023*. Nature Scot. Available from: <https://www.nature.scot/doc/proportion-scotlands-protected-sites-favourable-condition-2023> [Accessed 27th June 2023].
- Olofsson, P., Foody, G.M., Stehman, S.V. & Woodcock, C.E. (2013) Making better use of accuracy data in land change studies: estimating accuracy and area and quantifying uncertainty using stratified estimation. *Remote Sensing of Environment*, **129**, 122–131.
- ONS. (2022) *Countries (December 2022) boundaries GB BFC*. Available from: https://geoportal.statistics.gov.uk/datasets/c41e268a175f417eb7278dbd16b6e6d7_0/explore?location=55.066339%2C-2.950000%2C5.67 [Accessed 5th October 2022].
- Ordnance Survey. (2023) *OS Terrain 50 DTM*. Available from: <https://beta.ordnancesurvey.co.uk/products/os-terrain-50#get> [Accessed 5th October 2022].
- Pearce, I.S. & van der Wal, R. (2002) Effects of nitrogen deposition on growth and survival of montane *Racomitrium lanuginosum* heath. *Biological Conservation*, **104**(1), 83–89.
- Pearce-Higgins, J.W., Grant, M.C., Beale, C.M., Buchanan, G.M. & Sim, I.M.W. (2009) International importance and drivers of change of upland bird populations. In: Bonn, A. & Allott, T. (Eds.) *Drivers of environmental change in uplands*. Abingdon: Routledge, pp. 209–227.
- Pebesma, E. (2018) Simple features for R: standardized support for spatial vector data. *The R Journal*, **10**(1), 439–446. Available from: <https://doi.org/10.32614/RJ-2018-009>
- Planet Labs PBC. (2023) *PlanetScope*. Available from: <https://developers.planet.com/docs/data/planetscope/> [Accessed 29th August 2023].
- R Core Team. (2022) *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. Available from: <https://www.R-project.org/>
- Ramsey, D.S., Forsyth, D.M., Wright, E., McKay, M. & Westbrooke, I. (2019) Using propensity scores for causal inference in ecology: options, considerations, and a case study. *Methods in Ecology and Evolution*, **10**(3), 320–331.

- Raschka, S. (2014) An overview of general performance metrics of binary classifier systems. *arXiv preprint arXiv:1410.5330*.
- Robinson, M., Rodda, J.C. & Sutcliffe, J.V. (2013) Long-term environmental monitoring of the Plynlimon catchment study. *Transactions of the Institute of British Geographers*, **38**, 451–463. Available from: <https://doi.org/10.1111/j.1475-5661.2012.00534.x>
- RSPB. (2007) *The uplands: time to change?* Sandy, UK: RSPB. Available from: <https://www.rspb.org.uk/globalassets/downloads/documents/positions/climate-change/the-uplands---time-to-change.pdf>
- RStudio Team. (2022) *RStudio: Integrated development for R*. RStudio, PBC. Available from: <http://www.rstudio.com/>
- Sansom, A.L. (1999) Upland vegetation management: the impacts of overstocking. *Water Science and Technology*, **12**, 85–92. Available from: <https://doi.org/10.2166/wst.1999.0533>
- Scotland's Moorland Forum. (2017) *The Muirburn code – management of moorland by burning and cutting*. NatureScot and Scottish Government.
- Scottish Government. (2011) *The burn code*. Edinburgh: Scottish Government.
- Scottish Government. (2013) *Fire and rescue service wildfire operational guidance*. Available from: <https://www.gov.scot/publications/fire-rescue-service-wildfire-operational-guidance/pages/1/> [Accessed 31st August 2023].
- Scottish Government. (2019) *Grouse Moor management review group report to the Scottish Government*. Available from: <https://www.gov.scot/publications/grouse-moor-management-group-report-scottish-government/documents/> [Accessed 3rd April 2024].
- Scottish Government. (2021a) *Wildlife management: consultation, part 2: Muirburn*. Available from: <https://www.gov.scot/publications/wildlife-management-scotland-consultation/pages/5/> [Accessed 3rd April 2024].
- Scottish Government. (2021b) *Scotland habitat and land cover map – 2020*. Available from: <https://spatialdata.gov.scot/geonetwork/srv/api/records/88cea3bd-8679-48d8-8ffb-7d2f1182c175> [Accessed 15th September 2022].
- Scottish Natural Heritage. (2010) *Condition of Designated Sites*. Available from: <https://www.nature.scot/doc/archive/condition-designated-sites>
- Scottish Parliament. (2023) *Wildlife management and Muirburn (Scotland) Bill*. Available from: <https://www.parliament.scot/bills-and-laws/bills/wildlife-management-and-muirburn-scotland-bill> [Accessed 8th August 2023].
- Spracklen, B.D. & Spracklen, D.V. (2023) Assessment of peatland burning in Scotland during 1985–2022 using Landsat imagery. *Ecological Solutions and Evidence*, **4**(4), e12296.
- Story, M. & Congalton, R.G. (1986) Accuracy assessment: a user's perspective. *Photogrammetric Engineering and Remote Sensing*, **52**(3), 397–399.
- Tayside Biodiversity. (2015) *Tayside biodiversity action plan-Upland, Montane habitat*. Available from: <https://www.taysidebiodiversity.co.uk/wp-content/uploads/2015/01/Section2Upland2.pdf> [Accessed 8th August 2023].
- Thacker, J.I., Yallop, A.R. & Clutterbuck, B. (2015) *Burning in the English Uplands—A Review, Reconciliation and Comparison of Results of Natural England's Burn Monitoring: 2005–2014*. Improvement Programme for England's Natura 2000 Sites (IPENS), 055.
- Thompson, D.B.A., MacDonald, A.J., Marsden, J.H. & Galbraith, C.A. (1995) Upland heather moorland in Great Britain: a review of international importance, vegetation change and some objectives for nature conservation. *Biological Conservation*, **71**(2), 163–178.
- Thompson, P.S., Douglas, D.J., Hoccom, D.G., Knott, J., Roos, S. & Wilson, J.D. (2016) Environmental impacts of high-output driven shooting of Red Grouse *Lagopus lagopus scotica*. *Ibis*, **158**(2), 446–452.
- Tuanmu, M.N., Vina, A., Roloff, G.J., Liu, W., Ouyang, Z., Zhang, H. et al. (2011) Temporal transferability of wildlife habitat models: implications for habitat monitoring. *Journal of Biogeography*, **38**(8), 1510–1523.
- UK Government. (2021) *The heather and grass etc. burning (England) regulations 2021*. Available from: <https://www.legislation.gov.uk/uksi/2021/158/contents/made> [Accessed 3rd April 2024].
- Watts, C.D., Naden, P.S., Machell, J. & Banks, J. (2001) Long term variation in water colour from Yorkshire catchments. *Science of the Total Environment*, **278**(1–3), 57–72.
- Watts, S.H., Mardon, D.K., Mercer, C., Watson, D., Cole, H., Shaw, R.F. et al. (2022) Riding the elevator to extinction: disjunct arctic-alpine plants of open habitats decline as their more competitive neighbours expand. *Biological Conservation*, **272**, 109620.
- Welsh Assembly Government. (2008) *The heather and grass burning code for Wales 2008*. Available from: <https://www.gov.wales/sites/default/files/publications/2018-01/heather-and-grass-burning-code.pdf> [Accessed 23rd April 2023].
- Whitfield, D.P. (2002) Status of breeding dotterel *Charadrius morinellus* in Britain in 1999. *Bird Study*, **49**, 237–249.
- Worrall, F., Clay, G.D., Marrs, R. & Reed, M.S. (2010) *Impacts of burning management on peatlands*. Scientific review. IUCN Peatland Programme. Available from: https://www.iucn-uk-peatlandprogramme.org/sites/www.iucn-uk-peatlandprogramme.org/files/Review%20of%20Impacts%20of%20Burning_0.pdf [Accessed 8th August 2023].
- Yallop, A.R., Thacker, J.I., Thomas, G., Stephens, M., Clutterbuck, B., Brewer, T. et al. (2006) The extent and intensity of management burning in the English uplands. *Journal of Applied Ecology*, **43**, 1138–1148.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Data S1. Supplementary Methods.

Figure S1. Training point distribution. Left panel shows no burn point while the right-hand panel shows burn points from 2018/19 and 2019/20 burn seasons. Basemap is open terrain data from Open Streetmap.

Figure S2. Burn (red) and no burn (white) points for burn season 2019/20 are shown overlying imagery from 2019 (left) and 2020 (right). The inset maps show the broad location of the imagery within the UK along with the UK distribution of burn (red) and no burn (white) points.

Figure S3. Test 1-km grid square polygon locations. Basemap is open terrain data from Open Streetmap.

Figure S4. Prediction area of interest.

Figure S5. Random 1-km grid squares selected for manual mapping of burn scars for additional burn classification validation.

Figure S6. Classified burn pixels from the 2017/18 burn season. Due to the fine resolution at which burns are predicted (10-m pixels), burning is mapped here at 6.25-km² resolution as total number of pixels burned per square.

Figure S7. Classified burn pixels from the 2018/19 burn season per 6.25 km².

Figure S8. Classified burn pixels from the 2019/20 burn season per 6.25 km².

Figure S9. Classified burn pixels from the 2020/21 burn season per 6.25 km².

Figure S10. Classified burn pixels from the 2021/22 burn season per 6.25 km².

Figure S11. Boxplot of total HadUK 1 km gridded rainfall (mm) per 1 km² for the spring period of each year.

Figure S12. Boxplot of total HadUK 1 km gridded rainfall (mm) per 1 km² for the winter period (Oct–Dec) of each year.

Figure S13. Boxplot of total HadUK 1 km gridded rainfall (mm) per 1 km² for the burn season (Oct–Apr).

Figure S14. Boxplot of total HadUK 1 km gridded rainfall (mm) per 1 km² for the spring period of each year in England only.

Figure S15. Boxplot of total HadUK 1 km gridded rainfall (mm) per 1 km² for the winter period of each year in England only.

Figure S16. Boxplot of total HadUK 1 km gridded rainfall (mm) per 1 km² for the burn season (Oct–Apr) for England only.

Table S1. Polygon test data validation results for random points in burns.

Table S2. Spectral Indices and Spectral Indices differences calculated.

Table S3. Percentage of available pixels for classification in each burn season from 2017/18 to 2021/22.

Table S4. Summary balance statistics for variables of interest in all data (a) and balanced data (b) sets for England.

Table S5. Summary balance statistics for variables of interest in all data (a) and balanced data (b) sets for Scotland.

Table S6. Model structure including the temporal period of Spectral indices (SI) and Spectral Indices difference (dSI) generation, and the variables used in each of the six models.

Table S7. Model evaluation statistics for DNN1 and DNN2 in burn seasons 2017/18–2021/22.

Table S8. Confusion matrix for visually mapped polygons generated from 25 randomly selected 1-km² per burn season (100 km² in total) across the AOI.

Table S9. Confusion matrix for visually mapped polygons within selected 13 1-km² across the AOI that exceeded both 80% moorland cover and 80% burn percentage.

Table S10. Classified burn extents predicted from pixels that were available in all five burn seasons 2017/18–2021/22 (i.e. those pixels that are not obscured by cloud or snow in any year).

Table S11. Classified burn extents for valid data areas/pixels in all five burn seasons 2017–2022.

Table S12. Percentage of deep peat areas within our prediction area that have valid imagery in each burn season. These are the areas not lost to cloud cover or snow masking.

Table S13. Predicted burn extents in alpine (montane) habitats in each year.